

Evidence Gap Analysis: Domain Mismatch Between Ecological Carbon Sequestration Queries and High-Energy Physics Literature

Input Question

Where do we put all the excess carbon dioxide?

Domain

Ecology

Validation Status

needs_data

Problem Statement

The user inquiry seeks to identify viable storage mechanisms (sinks) for excess atmospheric carbon dioxide, specifically referencing geologic and biologic sequestration methods mentioned in a provided booklet excerpt. However, the strictly allowed evidence set consists exclusively of high-energy physics preprints discussing particle detection 'excesses' (e.g., diboson, gamma-ray, diphoton), which are semantically unrelated to ecological carbon cycles.

Rationale

A valid scientific response requires empirical evidence supporting specific sequestration mechanisms. The provided EvidenceCards (EV-Q108-696da4785392efdc7e5cd7ed, EV-Q108-47e1c9c77be956a0dd009e6c, EV-Q108-d9f75fc282c15b99a817a238, EV-Q108-076920795f5946afa98377d8) contain zero information regarding carbon dioxide, climate change, or environmental science. Therefore, the primary scientific task is to formally document this evidence gap and validate the domain mismatch to prevent hallucinated claims about carbon storage capacities or methods.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Current allowed evidence is insufficient to generate a valid scientific hypothesis regarding ecological carbon dioxide sequestration due to complete domain mismatch between the question (Ecology) and provided evidence (High-Energy Physics).
- **Mechanism**: The semantic retrieval system likely matched the keyword 'excess' from the user query ('excess carbon dioxide') with 'excess' in particle physics titles (e.g., 'diboson excess', 'Galactic Centre excess'), resulting in a set of evidence cards that contain zero information about carbon cycles, geologic formations, or biological sinks. Therefore, no causal mechanism linking CO₂ emissions to storage can be constructed from the

allowed IDs.

- ****Falsifiable Prediction****: If a comprehensive re-evaluation of the four allowed evidence IDs (EV-Q108-696da4785392efdc7e5cd7ed, EV-Q108-47e1c9c77be956a0dd009e6c, EV-Q108-d9f75fc282c15b99a817a238, EV-Q108-076920795f5946afa98377d8) reveals any mention of carbon dioxide, climate change, or ecological sequestration, this hypothesis of total irrelevance will be falsified.
- ****Required Observations****: Full-text verification of EV-Q108-696da4785392efdc7e5cd7ed confirming it discusses ATLAS diboson excess only. ; Full-text verification of EV-Q108-47e1c9c77be956a0dd009e6c confirming it discusses Galactic Centre gamma ray excess only. ; Full-text verification of EV-Q108-d9f75fc282c15b99a817a238 confirming it discusses 750GeV diphoton excess only. ; Full-text verification of EV-Q108-076920795f5946afa98377d8 confirming it discusses CDF Wjj excess only.
- ****Risk of Being Wrong****: Low risk. The quoted text snippets explicitly reference particle physics phenomena (gauge bosons, dark matter, scalar models, baryonic Z') with no ecological terminology. The primary risk is if the full text contains an unindexed interdisciplinary section, which is statistically negligible for these specific arXiv preprints.

Hypothesis 2

- ****Hypothesis****: Geologic carbon sequestration is the primary sink for excess anthropogenic CO₂ as stated in the booklet excerpt.
- ****Mechanism****: CO₂ is captured and injected into underground geologic formations where it is stored via structural trapping, residual trapping, solubility trapping, and mineral trapping.
- ****Falsifiable Prediction****: If valid ecological evidence were available, we would observe quantitative data supporting geologic storage capacity exceeding biologic storage; however, under current constraints, this hypothesis cannot be tested against allowed evidence.
- ****Required Observations****: Peer-reviewed studies on geologic CO₂ storage capacity. ; Comparative analysis of geologic vs. biologic sequestration rates.
- ****Risk of Being Wrong****: Extremely High / Certain Failure. This hypothesis relies entirely on the booklet excerpt without support from any allowed evidence ID. Per system rules, claims without evidence_id backing are prohibited. This hypothesis is generated solely to demonstrate the knowledge gap and must be rejected.

Technical Details

The recommended hypothesis posits that the provided evidence set is semantically irrelevant to the ecological question of carbon dioxide sequestration. The evidence cards (EV-Q108-696da4785392efdc7e5cd7ed, EV-Q108-47e1c9c77be956a0dd009e6c, EV-Q108-d9f75fc282c15b99a817a238, EV-Q108-076920795f5946afa98377d8) exclusively discuss high-energy physics phenomena (diboson excess, Galactic Centre gamma-ray excess, 750GeV diphoton excess, CDF Wjj excess). This experiment designs a verification protocol to confirm the absence of ecological keywords and concepts in these documents, thereby validating the 'insufficient_evidence' status and preventing hallucinated scientific claims.

Datasets

Source

```
```json
```

```
[
```

```
{
```

```
"id": "EV-Q108-696da4785392efdc7e5cd7ed",
```

```
"description": "Full text of arXiv:1506.06739 regarding ATLAS diboson excess."
```

```
},
```

```
{
```

```
"id": "EV-Q108-47e1c9c77be956a0dd009e6c",
```

```
"description": "Full text of arXiv:1404.4977 regarding Galactic Centre gamma ray excess."
```

```
},
```

```
{
```

```
"id": "EV-Q108-d9f75fc282c15b99a817a238",
```

```
"description": "Full text of arXiv:1512.06799 regarding 750GeV diphoton excess."
```

```
},
```

```
{
```

```
"id": "EV-Q108-076920795f5946afa98377d8",
```

```
"description": "Full text of arXiv:1104.1375 regarding CDF Wjj excess."
```

```
}
```

```
]
```

```
```
```

Target

```
```json
```

```
{
```

```
"description": "Binary classification output indicating presence/absence of ecological sequestration terminology in each document."
```

```
}
```

```
```
```

Paper Abstract

Background: The question of where to store excess atmospheric carbon dioxide is critical for climate mitigation, with potential solutions including geologic and biologic sequestration. Methods: We analyzed the four strictly allowed evidence cards provided for this query (EV-Q108-696da4785392efdc7e5cd7ed, EV-Q108-47e1c9c77be956a0dd009e6c, EV-Q108-d9f75fc282c15b99a817a238, EV-Q108-076920795f5946afa98377d8). Verification Protocol: Full-text semantic analysis was designed to detect ecological keywords versus high-energy physics terminology. Results: Pending execution of the verification script. Initial inspection of quoted texts confirms all sources relate to particle physics anomalies (diboson, gamma-ray, diphoton, dijet excesses) and contain no data on carbon cycles. Conclusion: The current evidence set is insufficient to answer the ecological question, necessitating the retrieval of domain-specific literature on carbon sinks.

Methods

Automated full-text semantic analysis and keyword extraction. The method involves parsing the PDF content of the four allowed Evidence IDs and searching for a predefined dictionary of ecological terms (e.g., 'carbon dioxide', 'sequestration', 'geologic storage', 'biological sink', 'climate change'). Simultaneously, it verifies the presence of high-energy physics terms (e.g., 'gauge boson', 'dark matter', 'diphoton', 'dijet') to confirm domain specificity. No machine learning model training is required; this is a deterministic verification of content relevance.

Experiments

Baselines

```
```json
```

```
[
```

```
"Random Keyword Match: Assuming a uniform distribution of scientific terms across all domains, calculating the probability of accidental ecological term occurrence in physics papers.",
```

```
"Domain Classifier Baseline: Using a pre-trained scientific document classifier (e.g., SciBERT) to predict the domain of each paper, expecting 'Physics' with >0.95 confidence."
```

```
]
```

```
```
```

Metrics

```
```json
```

```
[
```

```
"Term Presence Count: Number of occurrences of target ecological keywords per document.",
```

```
"Domain Confidence Score: Probability assigned to 'High Energy Physics' vs 'Ecology/Environmental Science' by the baseline classifier.",
```

```
"Relevance Verification Rate: Percentage of documents confirmed to contain zero ecological sequestration mechanisms."
```

]

...

## Ablation

Not applicable for deterministic text verification, but sensitivity analysis will be performed on the keyword dictionary size to ensure robustness against synonym variations (e.g., 'CO2' vs 'carbon dioxide').

## Validation Protocol

Cross-validation by manual inspection of abstracts and introductions of the four papers to confirm automated findings. If any ecological term is found, it must be contextualized to determine if it refers to the user's query or is a metaphorical/accidental usage.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- \*\*EV-Q108-696da4785392efdc7e5cd7ed\*\* · arxiv · arXiv:1506.06739
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1506.06739.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=bf6c47d376d6f0b1865635a5adfdaf04bca0ffee61087f2f20f1d35d0e07ee25
- \*\*EV-Q108-47e1c9c77be956a0dd009e6c\*\* · arxiv · arXiv:1404.4977
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1404.4977.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=2b2715fb13013b32f329a67eb2b31f7b12ee73d1dd1d072997cf4bb6b1f8c6f5
- \*\*EV-Q108-d9f75fc282c15b99a817a238\*\* · arxiv · arXiv:1512.06799
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1512.06799.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:2|section:page-2|paragraph:1;  
content\_sha256=c23a558a5c61a1755ce05df127618235962e72c8702736508cbe49c2573514d6
- \*\*EV-Q108-076920795f5946afa98377d8\*\* · arxiv · arXiv:1104.1375

- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1104.1375.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=9923a9f66d0667332fd23212fca0004d8fb57fee3275016e9e1de561f9c352dc

## Reviewer Comments

- The candidate hypothesis correctly identifies a complete domain mismatch between the user query (Ecology/Carbon Sequestration) and the allowed evidence (High-Energy Physics).
- The system successfully avoided hallucinating ecological facts from particle physics papers, adhering strictly to the 'insufficient\_evidence' protocol.
- Experiment design is appropriate for verifying the negative result (domain irrelevance) using deterministic text analysis rather than wet-lab or simulation methods.
- Results field is correctly marked as pending/not executed, avoiding fabrication of verification outcomes.
- All referenced evidence IDs exist in the allowed list and are accurately characterized as physics literature based on quoted text.

## Revision History

## Reproducibility Checklist

- Access to full-text PDFs of the four specified arXiv preprints via provided URLs.
- Defined list of ecological keywords for search.
- Defined list of high-energy physics keywords for confirmation.
- Script for PDF text extraction and string matching.
- Pre-trained SciBERT model or equivalent for domain classification baseline.